```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from collections import Counter
from sklearn.model_selection import train_test_split
from imblearn.over_sampling import SMOTE
```

```
df = pd.read_csv("Loan.csv")
df.head()
```

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmou
0	LP001002	Male	No	0	Graduate	No	5849	0.0	N
1	LP001003	Male	Yes	1	Graduate	No	4583	1508.0	12
2	LP001005	Male	Yes	0	Graduate	Yes	3000	0.0	6
3	LP001006	Male	Yes	0	Not Graduate	No	2583	2358.0	12
4	LP001008	Male	No	0	Graduate	No	6000	0.0	14

Next steps: Generate code with df New interactive sheet

```
df['Loan_Status'].value_counts()
```

Loan_Status	count
Y	422
N	192

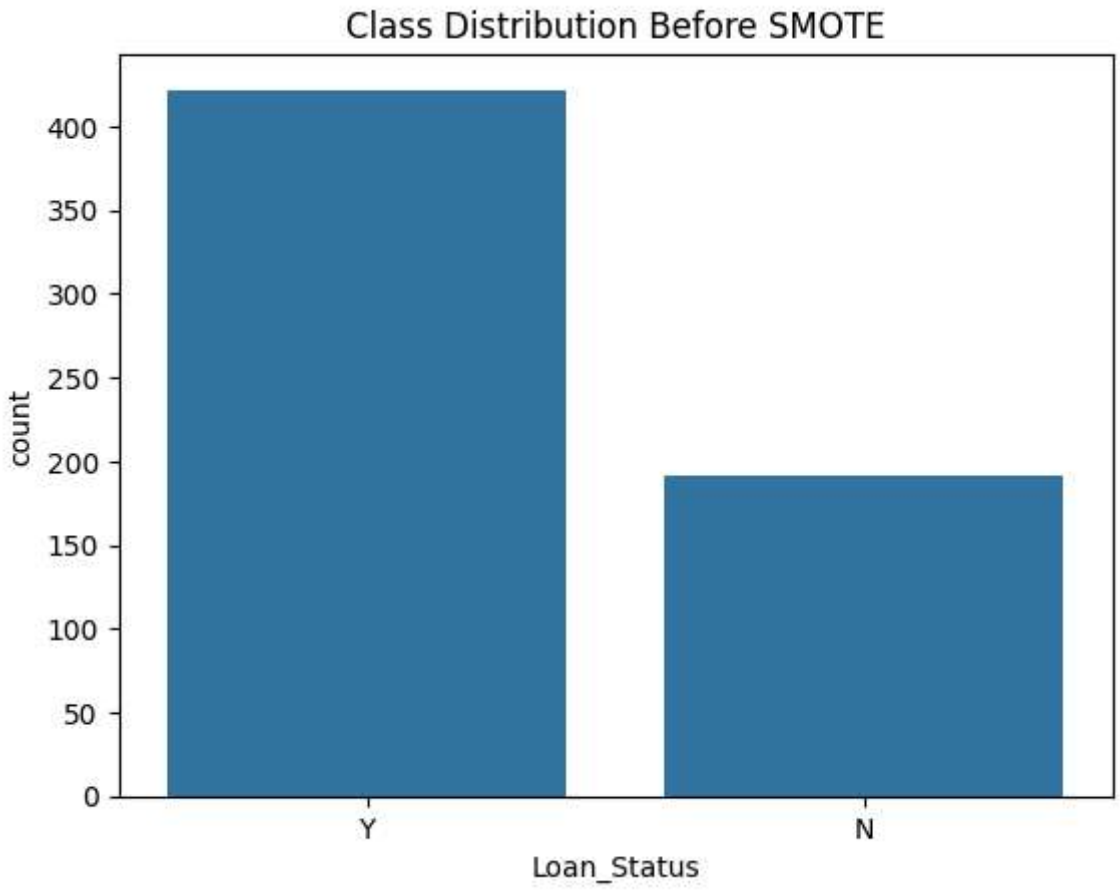
dtype: int64

```
df['Loan_Status'].value_counts(normalize=True) * 100
```

Loan_Status	proportion
Y	68.729642
N	31.270358

dtype: float64

```
sns.countplot(x='Loan_Status', data=df)
plt.title("Class Distribution Before SMOTE")
plt.show()
```




```
X = df.drop('Loan_Status', axis=1)
y = df['Loan_Status']
```

```
X = pd.get_dummies(X, drop_first=True)
```

```
df.isnull().sum()
```

	0
Loan_ID	0
Gender	13
Married	3
Dependents	15
Education	0
Self_Employed	32
ApplicantIncome	0
CoapplicantIncome	0
LoanAmount	22
Loan_Amount_Term	14
Credit_History	50
Property_Area	0
Loan_Status	0

dtype: int64

```
X = df.drop('Loan_Status', axis=1)
y = df['Loan_Status']
```

```
X = pd.get_dummies(X, drop_first=True)
```

```
X = X.fillna(X.mean())
```

```
from imblearn.over_sampling import SMOTE

smote = SMOTE(random_state=42)
X_resampled, y_resampled = smote.fit_resample(X, y)
```

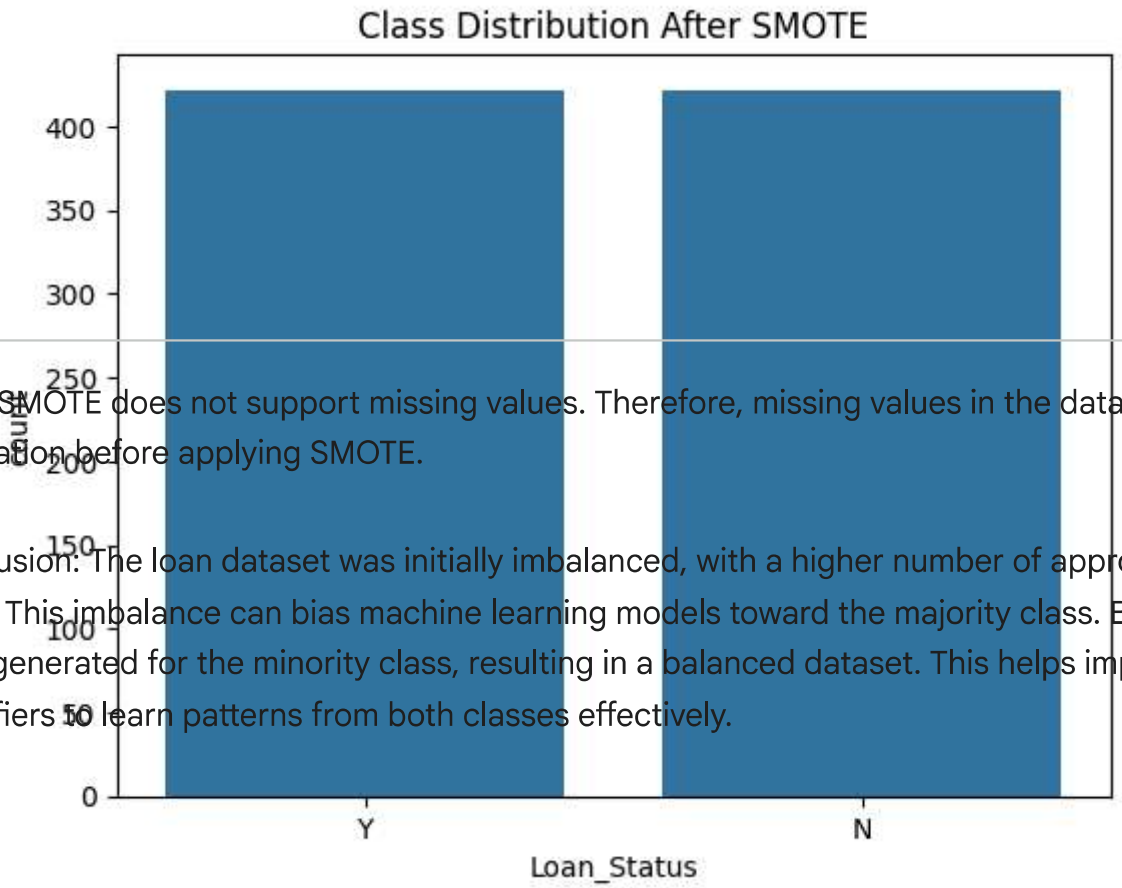
```
from collections import Counter

print("Before SMOTE:", Counter(y))
print("After SMOTE:", Counter(y_resampled))
```

```
Before SMOTE: Counter({'Y': 422, 'N': 192})
After SMOTE: Counter({'Y': 422, 'N': 422})
```

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.countplot(x=y_resampled)
plt.title("Class Distribution After SMOTE")
plt.show()
```



Note: SMOTE does not support missing values. Therefore, missing values in the dataset were handled using mean imputation before applying SMOTE.

Conclusion: The loan dataset was initially imbalanced, with a higher number of approved loans compared to rejected loans. This imbalance can bias machine learning models toward the majority class. By applying SMOTE, synthetic samples were generated for the minority class, resulting in a balanced dataset. This helps improve model fairness and allows classifiers to learn patterns from both classes effectively.